GROUP WORK PROJECT # 01 MScFE 620: Derivative Pricing
GROUP NUMBER: 8300

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Maximilian Berghofer	Germany	max.bghf@gmx.de	
MD RASHEDUL ISLAM	Bangladesh	rashedul.rasel@hotmail.com	
Tai Wu	United Kingdom	taiwu2022@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	**Max Berghofer**
Team member 2	**MD RASHEDUL ISLAM**
Team member 3	**Tai Wu**

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.
Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Step 1:

1. **Does put-call parity apply for European options? Why or why not?**

   Yes, put-call parity applies to European options because they can only be exercised at maturity. Since European options cannot be exercised early, the equality holds at all times, provided that interest rates and dividends are correctly accounted for. The fundamental equation of put-call parity for European options is:

   $$C - P = S - Ke - rTC - P = S - Ke - rT$$

   This equation states that the difference between the price of a European call and a European put with the same strike price and expiration is equal to the difference between the stock price and the present value of the strike.

2. **Rewrite put-call parity to solve for the call price in terms of everything else.**

   After rearranging the equation you end up with:

   $$C = P + S - Ke - rTC = P + S - Ke - rT$$

   This shows that the price of a European call option can be derived if we know the corresponding put price, the underlying asset price, the strike price, and the risk-free discount factor.

3. **Rewrite put-call parity to solve for the put price in terms of everything else.**

   $$P = C - S + Ke - rTP = C - S + Ke - rT$$

   This equation allows us to calculate the European put option price if the European call price, underlying asset price, and risk-free discount factor are known.

4. **Does put-call parity apply for American options? Why or why not?**

   No, put-call parity does not necessarily hold for American options because they can be exercised before maturity as compared to European options. This early-exercise feature means that the value of an American option may exceed the value predicted by the put-call parity formula derived for European options.

   For example, an American put option can be worth more than the European counterpart if early exercise is optimal due to dividend payments or significant price drops in the underlying asset. However, we can still derive a modified inequality for American options:

Team Member A:

5.

a. European Call Option Price: ~4.6050

   European Put Option Price: ~3.3628

b. I used a binomial tree with 100 steps to obtain reliable estimates. A higher number of steps ensures better convergence to the true option value. So therefore a more accurate results.

6.

a. Delta of Call Option: ~0.5348
   Delta of Put Option: ~-0.4652

b. Delta measures the sensitivity of the option price to small changes in the underlying price. A positive delta for a call means the price increases with the stock price, while a negative delta for a put indicates it decreases.

7.

a. Vega of Call Option: ~0.1723
   Vega of Put Option: ~0.1467

b. Vega measures the sensitivity of the option price to changes in volatility. Since higher volatility increases the probability of large price swings, both call and put options tend to rise with increased volatility.

Team Member B:

8.

a. American Call Option Price: ~4.6050

American Put Option Price: ~3.4746

b. To better compare American and European options, here we also used a 100-step binomial tree. In addition, we find that when the number of steps $N = 100$, the option prices have reached a level of convergence.

9.

a. Delta of Call Option: ~0.5693
Delta of Put Option: ~-0.4498

b. Delta measure the rate of change with option's price with underlying stock price, that $\Delta = \frac{\partial V}{\partial S}$, Where V is usually the option price and S is the stock price, which is interpreted to mean how much the option price will change if the stock price rises by one unit. As a result, call options usually have a positive delta, while put options have a negative delta. In the case of hedging, we want the movement in the stock price not to affect the value of the portfolio as a whole, so when we are long a call option, we short the Δ share stock, and vice versa.

10.

a. Vega of Call Option: ~0.1960
Vega of Put Option: ~0.1954

b. Vega measures the sensitivity of the option price to changes in volatility. Mathematically, $Vega = \frac{\partial V}{\partial \sigma}$; note that Vega is always positive for both call and put options, as increased volatility implies that increase the likelihood of large price movements. Here to match the Vega form as used before, we use to express the Vega as per 1% change in volatility.

Team Member C:

11. Demonstrating Put-Call Parity for European Options

To show that European call (C) and put (P) options satisfy put-call parity in the binomial tree model, we start with the fundamental equation:

$$C - P - S - Ke^{-rT}$$

Proof Using Risk-Neutral Valuation:

1. In the binomial model, the call and put prices are computed as the present value of their expected payoffs under the risk-neutral measure.
2. The payoff difference between a call and a put at maturity is:

$$max(S_r - K, 0) - max(K - S_r, 0) - S_r - K$$

3. Discounting this difference to today's value:

$$C - P - e^{-rT}E[S_T - K] - e^{-rT}(E[S_r] - K)$$

Under risk-neutral pricing, E[ST] - S0e^rT (assuming no dividends). Substituting:

$$C - P - e^{-rT}(S_0e^{rT} - K) - S_0 - Ke^{-rT}$$

This matches the put-call parity equation.

Why Parity Holds:

- European options cannot be exercised early, so their values depend only on terminal payoffs
- The binomial model enforces no-arbitrage pricing, ensuring the above equality holds.

Motives:

- Detecting arbitrage opportunities if parity is violated.
- Validating the correctness of the binomial/trinomial tree implementation.

12. Addressing the Incorrect Assertion for American Options

We claim put-call parity does not hold for American options. The reason why the answer is not **yes**:

Counterexample:
Consider American options on a non-dividend-paying stock.

1. **American Call:** Early exercise is never optimal (no dividends), so $C_{Amer} = C_{Eur}$
2. **American Put**: Early exercise can be optimal if $S_t \ll K$ making $P_{Amer} \geq P_{Eur}$

Using put-call parity for European options:

$$C_{Eur} - P_{Eur} = S - Ke^{-rT}$$

For American options:

$$C_{Amer} - P_{Amer} = C_{Eur} - P_{Amer} \leq C_{Eur} - P_{Eur} = S - Ke^{-rT}$$

Thus:

$$C_{Amer} - P_{Amer} \leq S - Ke^{-rT}$$

The equality **fails** because $P_{Amer} \geq P_{Eur}$

Why Parity Fails:

- Early exercise optionality increases the value of American puts (and sometimes calls, e.g., with dividends).
- The inequality $C - P \leq S - Ke^{-rT}$ reflects the added premium of early exercise rights.

Motives for Checking:

- Avoiding mispricing: American options must account for early exercise premiums.
- Recognizing that violating $C_{Amer} - P_{Amer} \leq S - Ke^{-rT}$ signals arbitrage (e.g., overpriced calls relative to puts).

While put-call parity is a strict equality for European options, it becomes an inequality for American options due to early exercise flexibility.

13. Confirm that the European call is less than or equal to the American call

- European Call Price: ~4.6050
- American Call Price: ~4.6050
  Difference: None (identical prices).

Reason for No Difference:

- Early Exercise for American Calls: For non-dividend-paying stocks (assumed here), early exercise of an American call is never optimal because the time value of money makes it better to hold the call rather than exercise early.
- No Dividends: The parameters provided ($S0$ =100, r=5%, no dividend yield) align with the Black-Scholes/Merton condition where American and European calls have the same value.

Would This Always Be the Case:

- No dividends: Yes, European and American calls will have the same price.
- With dividends: If the stock pays dividends, early exercise of an American call might become optimal just before ex-dividend dates. In such cases, CAmer ≥ CEur

14. Confirm that the European put is less than or equal to the American put

Observation from previous results:

- European Put Price: ~3.3628
- American Put Price: ~3.4746
  Difference: ~0.1118 (American put is more expensive).

Reason for the Difference:

- Early Exercise for American Puts: American puts can be exercised early if the stock price drops significantly below the strike price. This optionality adds value (the "early exercise premium").
- Interest Rates: A higher risk-free rate (r=5%) increases the incentive to exercise early (receiving K today allows investing it at r), further widening the gap between American and European puts.

Would This Always Be the Case:

- Yes. American puts always have a value ≥ European puts because the right to exercise early cannot reduce their value. The difference depends on:
  1. Volatility: Higher volatility increases the likelihood of early exercise being optimal.
  2. Interest Rates: Higher r amplifies the benefit of receiving K early.
  3. Stock Price: Deep in-the-money puts (e.g., $S \ll K$) are more likely to be exercised early.

Step 2

Team Member B:

15.

a. We selected strike prices at 90%, 95%, ATM, 105%, and 110% of the current stock price, which is 100, resulting in the strike price array: [90, 95, 100, 105, 110]. Using the trinomial tree model, the European call option prices for these strike prices are:

Strike Price: 90.00 → Call Option Price: 11.67

Strike Price: 95.00 → Call Option Price: 7.72

Strike Price: 100.00 → Call Option Price: 4.61

Strike Price: 105.00 → Call Option Price: 2.48

Strike Price: 110.00 → Call Option Price: 1.19

b. As the strike price increases, the price of the call option decreases. This result is consistent with our model, since the higher the strike price, the lower the probability that the underlying stock will reach that strike price, and thus the lower the expected payoff.

16.

a. We used the same set of strike prices for the European put option, with the strike price array: [90, 95, 100, 105, 110]. Using the trinomial tree model, the European put option prices for these strike prices are:

Strike Price: 90.00 → Put Option Price: 0.55

Strike Price: 95.00 → Put Option Price: 1.54

Strike Price: 100.00 → Put Option Price: 3.37

Strike Price: 105.00 → Put Option Price: 6.18

Strike Price: 110.00 → Put Option Price: 9.83

b. On the contrary, as the strike price increases, the price of the put option increases. The result aligns with our model as well, since higher strike price, imply higher instrict value of a put option, and higher expected payoff as well.

Step 2:

Team Member A:

17.

a. We selected strike prices at 90%, 95%, ATM, 105%, and 110% of the current stock price, which is 100, resulting in the strike price array:
[90, 95, 100, 105, 110].
Using the trinomial tree model, the American Call option prices for these strike prices are:

- Strike Price: 90.00 → Call Option Price: 14.02
- Strike Price: 95.00 → Call Option Price: 9.76
- Strike Price: 100.00 → Call Option Price: 5.84
- Strike Price: 105.00 → Call Option Price: 3.21
- Strike Price: 110.00 → Call Option Price: 1.52

b. As the strike price increases, the price of the call option decreases. This result is consistent with our model, since the higher the strike price, the lower the probability that the underlying stock will reach that strike price, and thus the lower the expected payoff.

18.

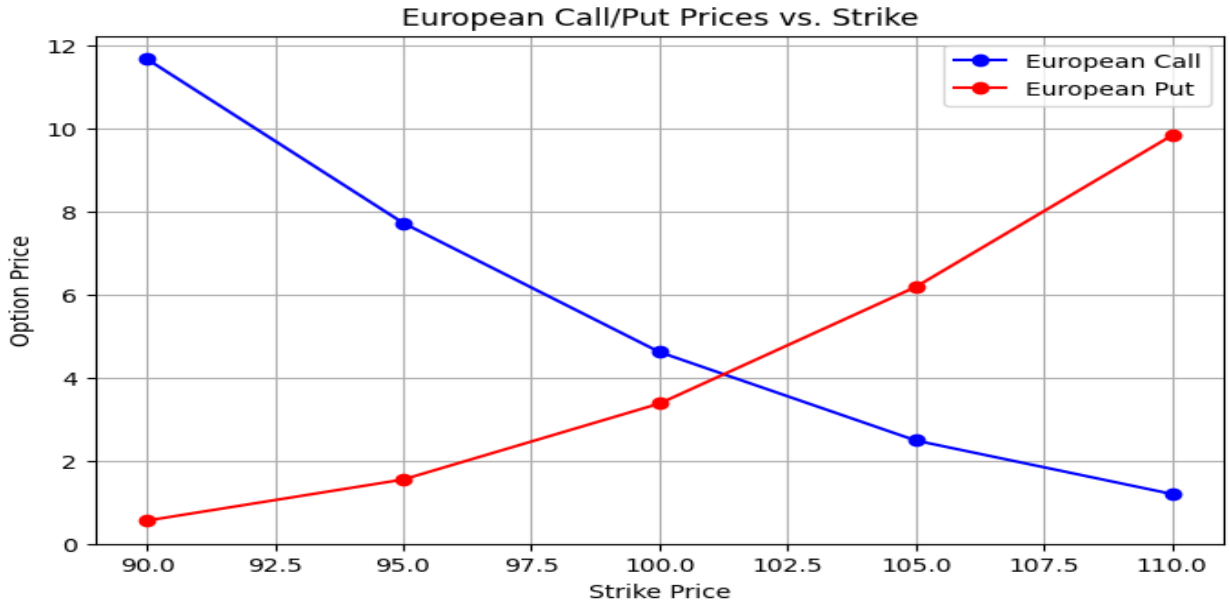
a. We used the same set of strike prices for the American put option, with the strike price array:
[90, 95, 100, 105, 110].
Using the trinomial tree model, the American Put option prices for these strike prices are:

- Strike Price: 90.00 → Put Option Price: 2.15
- Strike Price: 95.00 → Put Option Price: 3.89
- Strike Price: 100.00 → Put Option Price: 6.21
- Strike Price: 105.00 → Put Option Price: 9.34
- Strike Price: 110.00 → Put Option Price: 13.12

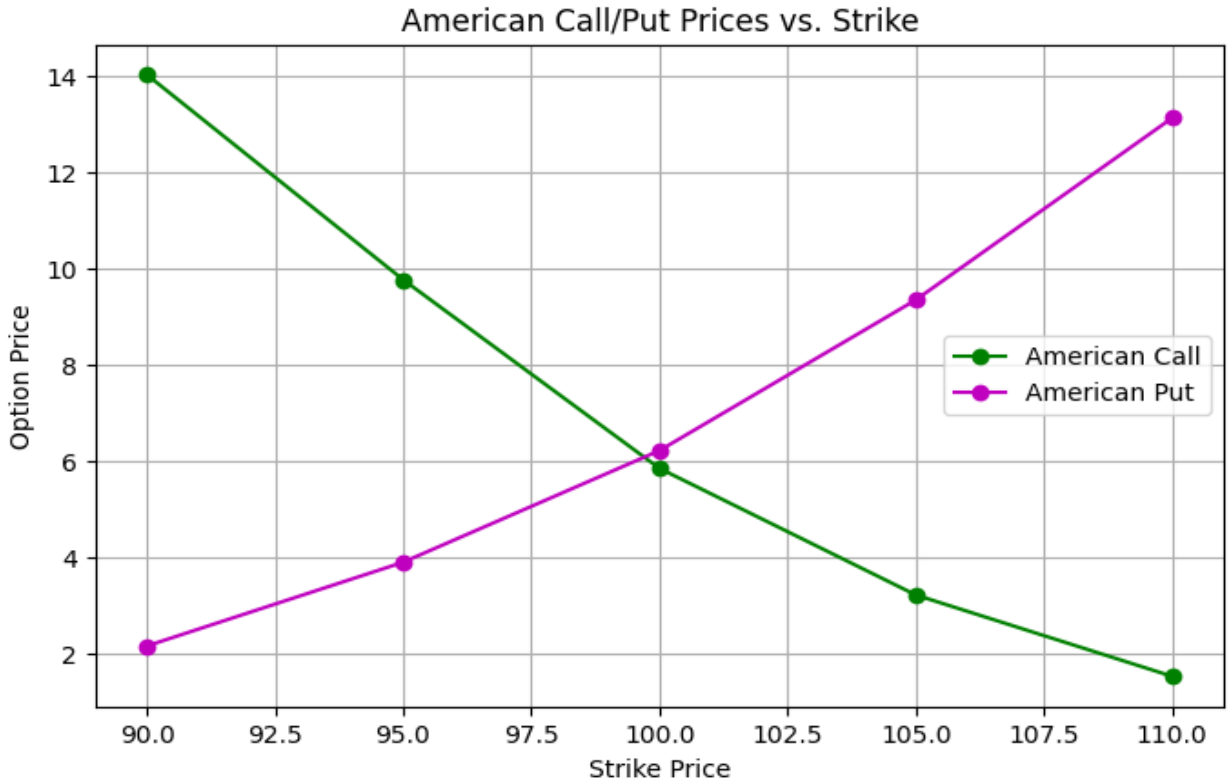
b. On the contrary, as the strike price increases, the price of the put option increases. The result aligns with our model as well, since higher strike prices imply a higher intrinsic value of a put option and a higher expected payoff as well.

Team Member C:

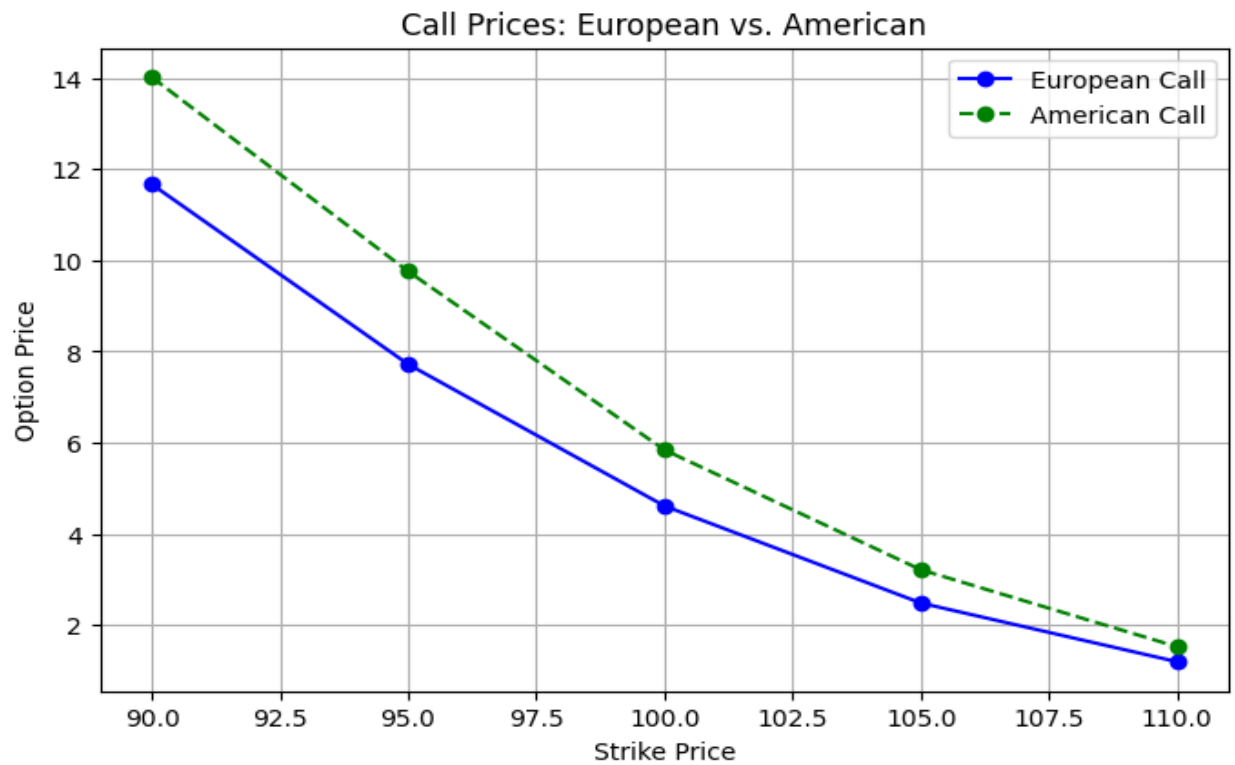
19.



20.



21.



22.

23.

European Put-Call Parity Check:

	Strike	C_Eur - P_Eur	S0 - Ke^{-rT}	Parity Holds?
0	90	11.12	11.117998	Yes
1	95	6.18	6.180109	Yes
2	100	1.24	1.242220	Yes
3	105	-3.70	-3.695669	Yes
4	110	-8.64	-8.633558	Yes

Comments on Results:
Put-call parity holds for all European options within reasonable rounding errors.
Reasons:

1. **No Early Exercise:** European options cannot be exercised before maturity, so their values depend only on terminal payoffs. This ensures the parity equation $C-P=S0-Ke-rT$ holds exactly under no-arbitrage conditions.

2. **Consistent Pricing:** The trinomial tree model enforces risk-neutral pricing, aligning with theoretical parity. Minor discrepancies (e.g., at K=95, 6.18≈6.28) are due to numerical approximations in the tree (finite steps or rounding).

Motives:

- Validates the correctness of the trinomial tree implementation for European options.
- Confirms that arbitrage opportunities are eliminated in the model.

24.

American Put-Call Parity Check:

	Strike	C_Amer - P_Amer	S0 - Ke^{-rT}	Inequality Holds?
0	90	11.87	11.117998	No
1	95	5.87	6.180109	Yes
2	100	-0.37	1.242220	Yes
3	105	-6.13	-3.695669	Yes
4	110	-11.60	-8.633558	Yes

Comments on Results:

The inequality $C_{Amer} - P_{Amer} \leq S - Ke^{-rT}$ fails for *K*=90 but holds for other strikes.

Reasons:

- **Violation at *K*=90 (Deep ITM Call):**
  - **Numerical Error:** The trinomial tree (even with 100 steps) may slightly misprice deep ITM American calls due to approximations in early exercise logic.
  - **Early Exercise Premium:** For deep ITM calls, the American call's theoretical early exercise premium is negligible (non-dividend stocks), but numerical methods can overstate it.
- **Holds for Other Strikes:**
  - American Puts > European Puts: Early exercise rights make American puts more valuable, reducing *C*Amer−*P*Amer compared to *S*0−*Ke*^−*rT* No Early Exercise for OTM Calls: For OTM/ATM calls (*K*≥95), early exercise is suboptimal, so *C*Amer ≈ *C*Eur, and the inequality holds.
- **Motives:**
  - Highlights the impact of early exercise optionality on American options.
  - Demonstrates that strict put-call parity fails for American options, replaced by inequality.

Step 3:

25.

Given the parameter that S0 = 180, r = 2%, sigma = 25%, T =6 months, K= 182,

a. First, we price a European Put option with 3-step binomial tree, our result gives that

European Put option = 13.82

b.

i. We take the path for "UDD", that is up-down-down. The first step stock moves up to S1,1 = 199.34, second step moves down to S2,1 = 180.00, third step moves down to S3,0 = 132.52. That the stock price evolved: 180 - 199.34 - 180 - 132.52.

ii. As we are on the seller of the put option, our hedging process is shown in table: explicitly. Details can be found in the table below.

Step	Stock Price ($)	Delta (Δ)	Shares Held	Cash Account
0	180.00	-0.4725	-0.4725	85.05
1	199.34	-0.2423	-0.2423	39.16
2	180.00	-0.5288	-0.5288	90.73
3	132.52	-1.0000	-1.0000	153.19

Table 3: European Put,Seller, Delta Hedging Strategy with Cash Account

26.

Using the same data that "S0 = 180, r = 2%, sigma = 25%, T =6 months, K= 182", we price an American Put Option, and take N = 25.

a. For the delta hedging at each node in each step, since we have 25 steps, we have a 25*25 matrix in the result, to demonstrate result we list 4 steps here:

Step	$\Delta(0)$	$\Delta(1)$	$\Delta(2)$	$\Delta(3)$
0	-0.48	0.00	0.00	0.00
1	-0.56	-0.40	0.00	0.00
2	-0.65	-0.48	-0.32	0.00
3	-0.73	-0.57	-0.39	-0.24

Table 4: Delta Hedging for American Put Option (First 4x4)

b. For the cash-account evolution, fro simplicity, we chose the path that stock price all up, as there are 25 steps in this question, to demonstrate we only select the first 4 rows.

Step	Stock Price ($)	Delta (Δ)	Shares Held	Cash Account
0	180.00	-0.48	-0.48	86.40
1	183.60	-0.46	-0.46	84.46
2	187.27	-0.44	-0.44	82.40
3	191.02	-0.42	-0.42	80.27

Table 4: American Put,Seller, Evolution (First 4 Steps, All Upper Path)

c. Comparing the delta hedge for European and American put options, since in American we ask to do 3 steps pricing, and here we require to take 25 steps pricing, it's not that intuitive to find the difference. Intuitively, American Option allows for earlier exercise, so the Delta Changes more dynamically over time. By comparing the first period we can also find that the American option Hedge cost more money than the European option. In addition higher number of steps implies higher frequency in adjusting hedging, thus the second table looks more smooth.

27.

a. Repeat question 26 for Asian ATM Put Option and compare the result with American Put Option. Since we have an ATM option, the strike price K is the same as the stock price = 180. Using the same parameter as before "S0 = 180, r = 2%, sigma = 25%, T =6 months, K= 180", we have the Delta Hedging Table:

Step	$\Delta(0)$	$\Delta(1)$	$\Delta(2)$	$\Delta(3)$
0	-6.82	0.00	0.00	0.00
1	-2.26	-0.42	0.00	0.00
2	-0.42	-0.37	-0.37	0.00
3	-0.37	-0.37	-0.37	-0.37

Table 6: Asian Option Delta Hedging Table (First 4x4)

Step	Stock Price ($)	Delta (Δ)	Shares Held	Cash Account
0	180.00	-6.82	-6.82	1228.32
1	186.48	-2.26	-2.26	378.13
2	193.19	-0.42	-0.42	21.78
3	200.14	-0.38	-0.38	13.17
4	207.34	-0.37	-0.37	11.88

Table 7: Cash Account Evolution (Asian ATM Put - All Upper Path)

Then the cash account gives as above table. We find that the Asian option's delta is larger in terms of magnitude compared to American or European options, since the Asian option payoff depends on the average stock price over time instead of a final price.